ACHYUTHAN RAGHAVAN

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SKILLS

- **Technical Skills:** KiCad, Altium Designer, 6-Layer PCB Design, BGA Escape Routing, Arduino, ESP32, Fusion 360, 3D Printing, Cabling & Grounding, Sensor Integration, Soldering, Micro-Soldering, Drones (quadcopter), ArduPilot / Mission Planner
- **Soft Skills:** Public Speaking, Team Leadership, Cross-functional Collaboration

INTERNSHIPS

Electronics Intern - Ecocipher, Gruner Renewable Energy — **Jun 2026 – Jul 2026**

- Built a 5 kg AUW fixed-wing drone from scratch for agricultural mapping — full system integration including frame assembly, motor/ESC selection, power wiring, and flight controller configuration.
- Configured and validated autonomous missions in ArduPilot/Mission Planner, executing waypoint-based survey flights over agricultural fields.
- Performed image processing on aerial imagery to support field analysis workflows.
- Debugged a custom 3-phase PMSM inverter, learning about Field Oriented Control (FOC) principles and micro-soldering.

Project Intern - IGCAR – Indira Gandhi Centre for Atomic Research, Kalpakkam — **May 2025 – Jul 2025**

- Developed a turbidity sensor for nuclear facility water monitoring; designed a custom ATmega2560 PCB, delivering a functional prototype within 8 weeks.
- Cut component costs ~35% and improved sensor accuracy by replacing an LDR sensor with a TSL235R light-to-frequency converter.
- Created full technical documentation (schematics, BOM) and 3D-printed a ruggedized housing in Fusion 360.

PROJECTS

30 × 30 mm 6-Layer STM32G491 BGA Sensor-Fusion PCB — **2026**

- Designed a compact 6-layer sensor platform in KiCad around an STM32G491 UFBGA64 MCU, integrating a 6-axis IMU, SPI NOR flash, USB 2.0, and an SWD debug interface.
- Completed manufacturing-ready layout with BGA escape routing, differential USB routing, and full DRC verification.

Rover Electrical & Controls System — **Apr 2024 – Present**

- Architected the full-stack electrical system for a Mars-analog rover, leading 5+ engineers to deliver a working prototype and cutting integration issues ~30%.
- Redesigned PCB layouts in Altium with a stackable architecture and integrated absolute/incremental encoders with actuator systems, enabling precise positional control, end-stop detection, and reduced external wiring complexity.

Solar Tracking Solar Panel — **Apr 2025**

- Leading a team of 6+ in developing an auto-aligning, servo-driven dual-axis solar tracker using light-sensor feedback.

ESP32-Based Game Boy — **2025**

- Built a handheld retro gaming console using ESP32, a TFT display, and a custom PCB with joystick/button input.

IoT-Based RC Cars — **2024**

- Built Wi-Fi-controlled RC cars on ESP32, designing motor driver circuitry and firmware for a custom web control interface.

LEADERSHIP

Electrical Subsystem Lead - Criss Robotics — **Apr 2025 – Mar 2026**

- Led and mentored 5+ electrical engineers, overseeing the full electronics stack of a Mars-analog rover from architecture through deployment.
- Aligned electrical design with firmware and structural constraints, driving key decisions on PCB design, power distribution, and actuator control.
- Onboarded and trained junior members on KiCad, Altium, and embedded systems as primary owner of electrical deliverables.
- Helped CRISS Robotics become the first Indian team invited to the Australian Rover Challenge (ARCh), demostrating as part of a group of 5 that represented the team in Australia.

President - Renewable Energy Club, BITS Pilani — **Apr 2025 – Apr 2026**

- Launched the first campus-wide Solar RC Racing competition and Cyclotron event, engaging students across departments.
- Managed a Solar Cooking event that raised over INR 10,000, donated entirely to Nirmaan, a student-run social welfare society.

COMPETITIONS

- 3rd Place, International Rover Design Challenge (IRDC) – 2025
- First Indian Team Invited to the Australian Rover Challenge (ARCh) – Part of a team of 5 that represented CRISS Robotics at ARCh, Australia
- Finalist, RoboFest Gujarat 5.0 – 2025
- Top 20 Teams Globally, International Rover Challenge, Goa – 2025
- Top 21 Teams Globally, International Rover Challenge, Manipal – 2026

EDUCATION

BITS Pilani, Pilani Campus — **2023 – Present**

BE Electronics and Communication Engineering

Courses: Digital Design, Microelectronics, Electrical Machines, Electronic Design, Control Systems, Signals and Systems, Microprocessors and Interfacing, Analog and Digital VLSI Design, Digital Signal Processing, Digital Image Processing, Microwave Engineering, Communication Systems

Master of Science in Physics (Integrated)

Courses: Physics of Semiconductor Devices, Electromagnetic Theory, Statistical Mechanics, Theory of Relativity, Quantum Mechanics, Computational Physics, Optics, Classical Mechanics, Nuclear and Particle Physics, Atomic and Molecular Physics, Solid State Physics